

Institute of Actuaries of India

Subject SA2 – Life Insurance

May 2023 Examination

INDICATIVE SOLUTION

Introduction

The indicative solution has been written by the Examiners with the aim of helping candidates. The solutions given are only indicative. It is realized that there could be other points as valid answers and examiner have given credit for any alternative approach or interpretation which they consider to be reasonable.

Solution 1:

i)

The Board approved policy for “Product Development and Pricing” should capture the following policies and procedures:

1. Applicability and Documentation

The policy should specify the various categories of products that can be developed by the Company. It should also specify the objectives / criteria to be adopted for selecting any new categories of products to be developed and procedures to be followed in developing the same.

It should also specify the details of the documentation to be maintained evidencing compliance with this policy. It should include the functions responsible for maintaining the documentation, the details to be maintained, the period for which they should be maintained etc.

2. Compliance with all applicable regulations and other policies

This may be an overarching requirement for the Company to comply with all applicable laws and regulatory requirements from time to time, as well as any other policies specified by the Board covering, for example:

- Distributor compensation
- Investments
- Audit
- Unit pricing
- Peer review

3. Personnel

The policy should identify the key personnel (e.g. CEO, CFO, CMO / CDO, AA, COO, CRO, etc.) who will be responsible for the key decisions related to product development and pricing matters. The policy may also mention about the responsibilities and ownership of any particular designated personnel.

4. Needs analysis

The policy should specify that any new product may be launched only in response to a specifically identified need for such a product in the market. This may be supported by any internal or external market analysis / market research.

5. Profitability and other criteria for shareholders’ financial outcomes

The policy should specify the key requirements of the Company related to profitability and other criteria. This may include, for example:

- a. Requirement for any product to be financially viable at each of the model point / cohort or portfolio level
- b. The financial viability should be measured by the following metrics:
 - Target minimum new business margins (NBM) on a traditional pricing basis (or on a market consistent pricing basis);
 - Target minimum shareholder IRRs (wherever it exists);
 - Maximum Discounted Payback Period;
 - Maximum New Business Strains etc.

The policy should specify the acceptable range for each of the above matrices for all major lines of businesses at the cohort or portfolio level.

- c. Framework for specifying various pricing assumptions, including the Risk Discount Rate (RDR) to be adopted in case of the traditional pricing basis.

- d. Procedures to be adopted in special circumstances requiring exceptions to the above requirements and procedures to be adopted in seeking approval from the Board for such exceptions.

6. Risk management

The policy may include requirements for assessing various risks arising and the Company's approach in mitigating / managing / limiting the same. This may cover:

- Need to carry out sensitivity analysis and scenario testing due to variation in demographic and economic assumptions on the various shareholder financial outcomes.
- Limits on maximum acceptable risks, specified through maximum incremental capital requirements, minimum reinsurance requirements, etc.
- Circumstances requiring any hedging arrangements to be put in place etc.

7. Requirements for policyholders' financial outcomes

This may include having an internal framework for Treating Customer's Fairly (TCF); target minimum policyholder IRRs; target minimum ratio of 'insurance' benefits to total premiums charged; basis for calculation of surrender values etc.

8. Ownership and approval process

The policy should have clearly defined roles and appropriate levels of delegation for ownership and approval of each material aspect of the development and pricing stages. This may cover, for example, the following roles and ownerships:

a. Product Development / Marketing

- Ensuring the consistency of all documents such as the 'File and Use' document, the product specifications, reinsurance treaty, prospectus / application forms, pricing models used, sales illustrations, product brochure and marketing material on the website etc.
- Adherence to 'New Product Readiness' process
- Adherence to 'Operational Governance' process
- Development of marketing / training materials

b. Actuarial

- Pricing bases are set in line with the Company's own experience (if credible) or based on the industry / reinsurers' experience (otherwise), to the extent applicable.
- Overall responsibility of Actuarial Reporting which includes readiness of the actuarial projection models, testing of data fields for financial projections etc.
- Finalization of reinsurance rates.
- Configuration and testing of premium rates, surrender rates etc. in the admin system.

c. Legal

- Drafting of the policy contract wording and ensuring compliance with the applicable insurance laws and regulations

d. Information Technology

- Ensuring the system readiness across various systems including those for underwriting, sales illustrations, policy admin, reinsurance and claims management etc.
- Review of test scenarios and comprehensiveness of the User Acceptance Testing (UAT) execution

9. Procedures and criteria for withdrawal / re-pricing of product

The policy should include the criteria to be applied and procedures to be followed for withdrawal or re-pricing of existing insurance products. This may require the Company to monitor emerging experience under all the

products and justify any re-pricing / modification only based on credibility of emerging experience and / or due to strong market indicators or regulatory changes justifying such modification/re-pricing.

[2 marks for each of the points appropriately described, Max 18]

ii) Following are the key risks that could have significant contribution towards the new Risk Based Capital (RBC) requirement:

1. Unit-linked products:

a. Lapse risk:

- Given that the acquisition expenses and commissions are typically front loaded (and which are expected to be recovered through various charges over the tenor of the policy) and given the regulatory limit on surrender penalties, unit-linked products typically have a significant lapse risk.
- For RBC assessment, the contribution from lapse risk is calculated as highest of the three scenarios - lapse 'up', lapse 'down' and mass lapse risks. Depending on the quantum of the factor specified in the stress scenario, mass lapse risk may be the most onerous of them.

b. Market risks:

- The movement in fund value under the stress scenarios of a fall in equity values or increase in interest rates would result in a reduction in future fund management charges required to recover the various expenses and commission outgoes. Such asset movements may also increase the level of guarantees (if offered).
- Consequently, for RBC assessment, the various market risks may be another key contributor to the capital requirements.

c. Operational risk:

- Considering the complexities involved in administration of the unit-linked products (e.g. estimation of daily unit prices; daily matching of fund balances with units records etc.), there are significant operational risks. However, there is no single scientific method to estimate operational risk capital requirements. Depending upon the approach / formula adopted to estimate this, operational risk could be another major contributor to the capital requirements.

2. Non-participating savings products:

a. Interest rate risk:

- Considering the likely mismatch between the asset and liability durations (asset durations likely to be shorter than the liability durations), interest rate 'down' scenario is expected to be more onerous for these products, contributing significantly to the capital requirements.
- To the extent the Company has hedged the interest rate risk through third party products such as Forward Rate Agreements (FRA), Interest Rate Swaps (IRS) and Interest Rate Futures (IRF), although the interest rate 'down' scenario may not lead to significant capital requirements, the Company may be required to provide capital to meet the counterparty default risk and concentration risks.

b. Lapse risk:

- Give the typical front loading of acquisition expenses and commissions, and considering the regulatory requirements to provide minimum guaranteed surrender values (GSV), these products are subject to a significant lapse risk. Depending on the stress factor specified, a mass lapse scenario may be a major contributor to the capital requirements.

3. Non-participating protection products:

a. Mortality and Catastrophe risks:

- Given the high level of 'protection' cover provided for relatively low level of premiums charged, the main risks for these products are the mortality and catastrophe risks, contributing significantly to the capital requirements.
- To the extent the Company has reinsured these risks, although the capital requirements arising from these risks may be lowered, the Company may be required to provide capital to meet the counterparty default risk.

b. Expense risk:

- Given the relatively low levels of premiums charged, the proportionate contribution of expenses in the premiums charged is typically very high. As a result, an increase in expenses is typically a significant risk for these products. Depending on the stress factor specified, this may have a significant contribution to the capital requirements.

c. Lapse risk:

- Given the front loading of acquisition expenses and commissions which are recovered over the longer term through typically higher profit loadings in these products, they are exposed to significant lapse risk and may lead to a significant contribution to the capital requirements.

[1 mark for each of the risks correctly identified with reasons, Max 8]

In the current Factors Based solvency regime, the capital requirement (i.e. the Required Solvency Margin of RSM) is derived indirectly – as the sum of a specified percentage of reserves and a specified percentage of ‘amount at risk’. Consequently, the allowance for various risks is also reflected indirectly – in either how the reserves are calculated or how the various percentages are specified.

[1 mark for identifying this at a high level]

1. **Unit-linked products:**a. Lapse risk:

- In setting the margins for adverse deviations (MAD) to be used in the reserving bases, the Actuarial Practice Standard 7 (APS7) specifies a minimum adverse scenario of (+/-) 20% of the best estimate lapse assumptions to be used.
- Also, any non-unit reserves are required to be zeroised and therefore, the Company is not allowed to take credit for higher charges in the future, thereby adding a level of prudence.

b. Market risks:

- This is allowed through the minimum adverse scenarios to be tested as per APS7, whilst setting the MADs for the interest rates (i.e. future fund growth rate and discount rate assumptions) used in reserves calculation.
- The minimum adverse scenario specified reflects an immediate increase / decrease in the best estimate assumption, by 10% of the current yield on 10-year Gilts for the next five years, followed by a further increase / decrease by 10% of the current yields on 10-year Gilts.
- Given that the APS7 requirements are considered to be ‘minimum’ adverse scenarios to be tested, there is an expectation that to provide for any investment guarantees, whilst setting the MADs, the Company would allow for a more adverse scenario than that specified in APS7. Alternatively, the Company may also provide for any guarantees explicitly through a stochastic approach, as specified in Guidance Note 22 (GN22).

c. Operational risk:

- The current regulatory regime for reserving and solvency capital assessment doesn’t mandate companies to explicitly hold capital for any operational risks. Such risks may be allowed for implicitly through additional margins in the valuation bases.

2. **Non-participating savings products:**a. Interest rate risk:

- This is reflected in the reserves through the application of MAD in the valuation interest rate assumption as required in APS7. The minimum adverse scenario specified reflects an immediate increase / decrease in the best estimate assumption, by 10% of the current yield on 10-year Gilts for the next five years, followed by a further increase / decrease by 10% of the current yields on 10-year Gilts.
- In case the Company has hedged its interest rate risk through derivatives, the credit for the same is reflected in the valuation interest rate assumption after allowing for the secondary risks associated with the hedging, such as the counterparty default risk etc.

b. Lapse risk:

- In setting the margins for adverse deviations (MAD) to be used in the reserving bases, the Actuarial Practice Standard 7 (APS7) specifies a minimum adverse scenario of a (+/-) 20% of the best estimate lapse assumptions to be used.
- The reserves are also required to be floored to the maximum of zero, the Guaranteed Surrender Value or the Special Surrender Value. This indirectly provides for a high level of prudence towards the lapse risks.

3. Non-participating protection products:a. Mortality and Catastrophe risks:

- The reserving assumption is required to be set based on a MAD, set by considering a minimum adverse scenario of 10% worsening of the 'best estimate' mortality assumption.
- Although the current regulatory regime allows reserving to reflect credit for any reinsurance arrangements, whilst assessing the RSM, the regulations specify maximum limits for any such reinsurance credits. For the component of the RSM that is based on reserves, the maximum credit allowed (reflecting 1 minus the ratio of reserves after reinsurance to reserves before reinsurance) is 15%, whereas for the component of RSM that is based on 'amount at risk', the maximum credit allowed (reflecting 1 minus the ratio of 'amount at risk' after reinsurance to that before reinsurance) is 50%.
- Although not explicitly required in the regulations, insurers may allow for the risk of default by a reinsurer whilst considering any credit for reinsurance cash-flows in the calculation of reserves or solvency capital.
- There is no explicit provision in the current regulatory regime to allow for the Catastrophe risk.

b. Expense risk:

- The allowance for expense risk is reflected by assuming a MAD over the 'best estimate' expense assumption in the calculation of reserves. The APS7 specifies a 10% of 'best estimate' assumption as the minimum adverse scenario for setting expense assumption.

c. Lapse risk:

- In setting the margins for adverse deviations (MAD) to be used in the reserving bases, the Actuarial Practice Standard 7 (APS7) specifies a minimum adverse scenario of (+/-) 20% of the best estimate lapse assumptions to be used.
- Also, the reserves are required to be zeroised and therefore, the Company is not allowed to take credit for future positive cash-flows (arising due to the high profit loadings), thereby adding a level of prudence.
- If the Company offers surrender values (e.g. in single premium or in longer term protection policies), the reserves are required to be set to the higher of the Guaranteed Surrender Value (GSV) or the Special Surrender Value (SSV).

[1 mark for appropriately describing how each of the risks are reflected, Max 8]

[Max 17]

iii) Reasons for increase in solvency ratio

The solvency ratio is the ratio of 'Available Solvency Margin' (ASM) and the 'Required Solvency Margin' (RSM). In the proposed RBC regime, the RSM may be referred to as 'Solvency Capital Requirement' (SCR). The estimation of RSM (or SCR) and ASM in both the capital assessment regimes may be very different.

A higher ASM and a lower SCR in the proposed RBC regime may lead to a higher solvency ratio than that in the current solvency regime.

[1 mark for identifying the reason at a high level]

Given below are the possible reasons for differences in the estimation of the ASM and RSM (or SCR), leading to an increase in the solvency ratio in the proposed RBC regime:

1. Differences in ASM calculation:

The ASM in the proposed RBC framework may be calculated as the market value of assets less the 'Best Estimate Liabilities' (BEL) less 'Risk Margin' (RM) whereas, in the current solvency regime, the ASM is calculated as the book value of assets (or market value for equities and derivatives, if lower than the book value) less the reserves. The possible differences in the estimation of each of these are:

a. Admissibility of assets

There may be differences in the two approaches in the admissibility of assets for the purpose of ASM calculation. For example, certain assets such as agents' balances and premiums outstanding for more than 30 days are not admissible for solvency purposes in the current regime. The treatment for admissibility of such assets in the proposed RBC regime may be different. If such assets are admissible, it may increase the ASM in the proposed RBC regime.

[1 mark for identifying with example]

b. Assets valuation

The unrealized gains/losses on fixed income securities may result in material difference in the book and market values of these assets. A fall in the interest rates may lead to a significant unrealized gain on such assets and may result in higher ASM in the proposed RBC regime.

Any changes in the credit spreads on the corporate bonds may also result in differences in the book and market values of such assets. A fall in credit spreads may lead to higher ASM in the proposed RBC regime.

Furthermore, any unrealized gains on equity/derivatives may also result in higher value of these assets, leading to a higher ASM in the proposed RBC regime as compared to that in the current solvency regime.

[2 marks for identifying with example]

c. Liabilities valuation

Considering that in the current solvency regime, the assets are valued on a book value basis, in the liability valuation too, whilst setting the valuation interest rate, the Company would consider the book yields on existing assets along with market yields for future investments. Such 'best estimate' interest rates are then adjusted for MADs to allow for prudence.

On the other hand, in the proposed RBC regime, the liability valuation may be based on fair market value, consistent with the market valuation of the assets.

Depending on the credit for risk margins reflected in the valuation interest rate used in the current solvency regime, the possible use of risk-free interest rates in the valuation of non-participating liabilities in the proposed RBC regime may lead to a higher overall liability valuation in the RBC regime.

On the other hand, the overall bases for the valuation of liabilities in the current solvency regime may be more prudent as compared to what may be prescribed in the proposed RBC regime due to:

- Differences in the MADs in demographic assumptions as per APS7 as compared to the approach that may be adopted in setting the RMs in the proposed RBC regime
- Artificial flooring to the reserves (such as zeroization or Special Surrender Values) in the current solvency regime may not apply to the liability valuation in the proposed RBC regime.

On balance, a lower overall liability valuation in the proposed RBC regime may lead to a higher ASM.

[3 marks for identifying with examples]

2. Differences in RSM (or SCR) calculation:

a. Difference in approach

- In the current solvency regime, The RSM is calculated based on a factor-based approach, wherein the factors (varying by lines of business) prescribed in the regulations get applied to the reserves and the 'sum at risk'.
- As these factors are prescribed in the regulations, they may not reflect the specific risks that the Company may be exposed to. They also lead to certain anomalies – e.g. a more prudent reserving bases may lead to a higher RSM and vice versa.
- Whereas in the proposed RBC framework, the companies are required to calibrate their own stress factors reflecting the risks underlying their businesses. Consequently, if the Company has better risk management techniques – e.g. better underwriting standards, a robust reinsurance arrangement, hedging of its interest rate risks in the non-participating savings portfolio, strict control on expenses / outsourcing contracts with vendors to lower expenses etc. – it may be able to reflect the same in calibrating its own – lower - stress factors. This may lead to a lower SCR requirement in the proposed RBC regime.

[2]

b. Difference in allowance for risk mitigation techniques

- In the current solvency regime, for the estimation of RSM, the credit allowed for the risk mitigation technique through reinsurance is restricted (only up to 15% credit on the 'reserves' component and 50% credit on the 'sum at risk' component), whereas in the proposed RBC regime, the Company may be able to take full credit for any reinsurance arrangement. This may result in a lower SCR in the proposed RBC regime.

[1]

c. Diversification benefit

- In the proposed RBC regime, the Company may be allowed to take credit for diversification benefit - which assumes that not all the risks are going to materialize at the same time. The current solvency regime doesn't provide for any such diversification benefit. Consequently, the SCR in the proposed RBC regime may be lower than that in the current solvency regime.

[1]

Reasons for increase in the sensitivity of the solvency ratio to changes in interest rates

A change in the interest rates impacts the valuation of assets and liabilities differently between the existing solvency regime and the proposed RBC regime, leading to a differing level of sensitivity of the resulting solvency ratios. These are discussed below:

1. Existing solvency regime

- a. The assets (specifically fixed interest securities) backing the non-unit liabilities are valued at amortized cost and hence any change in the interest rates would have no impact on the valuation of the assets.

The liabilities may change due to changes in the interest rates, given that the future new money from the existing business will now be invested at a different yield, resulting in an overall change in the 'best estimate' of expected portfolio yield based on which the valuation interest rate is derived. However, given that the valuation interest rate may change only to the extent of changes in future new money yields, the overall change in the valuation interest rate may be subdued.

- b. Given the relatively insensitive nature of the valuation of assets and liabilities to changes in interest rates, the resulting ASM and RSM (and therefore the resulting solvency ratio) of the Company may also be less sensitivity to changes in interest rates.

2. Proposed RBC regime

- a. The valuation of both assets and liabilities are likely to be based on market consistent valuation methodology. Thus, any change in the interest rates may have a material impact on the valuation of assets as well as the liabilities.
- b. The magnitude of the impact on the valuation of assets and liabilities is dependent on the differences in their respective durations. The Company has been selling non-participating savings and protection business. It is possible that the asset duration may be shorter (potentially significantly so) than the liability duration. This may lead to the resulting asset valuation being less sensitive than the liability valuation to changes in interest rates.
- c. To a certain extent, the impact of this mismatch in duration of the assets and the liabilities in the policyholders' funds may be muted by the movement in the valuation of assets backing the shareholders' fund, given that there may not be any corresponding liabilities in that fund. Nonetheless, if the overall size of the policyholders' funds is significantly larger than the shareholders' fund, the Company level sensitivity of the solvency ratio may still be influenced by the mismatch between the asset and liability durations in the policyholders' funds.
- d. Given the relatively sensitive nature of the valuation of assets and liabilities to changes in interest rates, the resulting ASM and RSM (and therefore the resulting solvency ratio) of the Company may also be more sensitivity to changes in interest rates.

[4 marks for identifying reasons for sensitivities separately in the two regimes]

[Max 15]

[50 Marks]

Solution 2:

- i) The different consumer needs that are catered to in these two products are discussed below:

1. Participating deferred annuity with guaranteed annuity rates (separate for standard and impaired lives)

- a. **Retirement income** – As this is a participating product, the value of the policy is expected to increase over the deferment period as the insurer earns profits and distributes the same in the form of bonuses. This would allow the policyholder to save for retirement:
 - Either through regular / limited premium payments,
 - Or through single premium / recurrent single premium contract.

The limited premium / recurrent single premium version may allow the policyholder the flexibility to not commit premium payments for the longer term, catering to such policyholders who do not have a stable income / savings capacity.

- b. **Varying needs during retirement** - If the annuities offered provide for different options, they may cater to different needs of the policyholder. For example:
 - Joint life / last survivor annuity → Providing regular income needs of the surviving spouse
 - Increasing annuity → Protection against inflation
 - Annuity guaranteed for certain period and for life thereafter → Certainty of payment for a minimum period (e.g. if there are any known regular outgoes expected – e.g. regular loan repayments)
 - Return of notional purchase price upon death during retirement → Lump-sum cash needs of family members
- c. **Meeting any additional expenses for impaired lives** – Given the separate rates offered for impaired lives, those policyholders expecting a shorter life-span (e.g. due to existing medical conditions) may be able to enjoy higher income during retirement, allowing them to meet any additional expenses.
- d. **Lump-sum cash needs at retirement** – At retirement / vesting, it may allow the policyholder to take a certain amount as a lump sum (i.e. commute a portion of the annuity payable) and receive the remaining amount in the form of an annuity. This would allow the policyholder to meet any immediate cash needs at retirement / vesting (e.g. to pay off any outstanding loan etc.) and still receive a regular income to support the ongoing needs during retirement.

- e. **Tax planning** - The lump sum benefit (commutation) at retirement / vesting could be tax-free. This may allow the policyholder to defer tax payments until retirement and save a portion of taxes that would have otherwise become payable on annuity payments.
- f. **Protection from low interest rates** - As the annuity rates are guarantees at inception of the contract, it may help the policyholder appropriately plan for the future income needs in retirement. Even if the interest rates at the time of retirement / vesting are significantly low (which would have otherwise warranted low annuity rates then), the policyholder will not be subjected to undue hardship given the guaranteed annuity rates provided.
- g. **Liquidity during deferment period** - During the deferment period, the product may offer surrender value (albeit with certain tax implications) and allow various options (lump-sum / annuitization etc.). This may provide liquidity, if desired, by the policyholder.
- h. **Tax efficient estate transfer** – Upon death of the policyholder during deferment, the product would provide tax-free death benefits to be received in lump-sum. This would allow a tax efficient estate transfer.
[1 mark for each point explaining the specific needs linked to the feature of the product, Max 4]

2. Non-participating life annuity payable upon diagnosis of critical illness (CI) or inability to perform the specified number of activities of daily living (ADL)

- a. **Income replacement** – Given that the annuity becomes payable upon diagnosis of the CIs or upon inability to perform ADLs, the main needs this product would fulfill is that of income replacement when the policyholder may be unable to earn any income due to his medical condition.

Given that in this product, cash is paid out without linking it to the actual cost of medical treatment, it can be used for any purpose, like meeting expenses related to children's education, household expenses, repayment of mortgages etc.

- b. **Higher level of income to meet expenses of an impaired life** - For the annuity to commence, given that the policyholder will be considered as a significantly 'impaired' life with a shorter life-span, the resulting annuity rates may be significantly higher than those offered for standard annuity products. This may also help the policyholder meet any additional expenses related to his medical condition and / or nursing support provided.
- c. **Protection from low interest rates** - As this is a non-participating plan and the annuity rates are guaranteed, the policyholder may be in a better position to plan any financial needs during his potential medical condition in the future. Even if the interest rates at the time of diagnosis of CI or inability to perform ADLs are significantly low, the policyholder will not be subjected to undue hardship given the guaranteed annuity rates provided.
- d. **Needs during claims payment period** - If the annuity offered provides for different options, they may cater to different needs of the policyholder. For example:
 - Increasing annuity → Protection against inflation in the future especially as the policyholder may continue to live for several years even after being diagnosed with a CI
 - Death benefit in the form of return of balance of the purchase price / premiums paid → Lump-sum cash needs of family members

[1 mark for each point explaining the specific needs linked to the feature of the product, Max 4]

[Max 10]

ii)

1. The key risks faced by the Company in these two products are discussed below:

Participating deferred annuity with guaranteed annuity rates

a. Market risks

- The main risk to the Company is lower than expected investment returns on its assets.
- During the deferment period, the participating nature of the product would mean that the Company may be able to adjust future bonuses in response to a lower-than-expected investment return. To the extent the Company does not create onerous guarantees through high bonus declarations early on, this risk may be relatively contained.
- However, as the annuity rates are guaranteed at the inception of the contract, the Company is subjected to significant investment risk at the time of vesting of the annuity benefit given that the interest rates at the time of vesting of annuity may be significantly lower than those reflected in the guaranteed annuity rates offered.
- If the product allows vesting of annuities at a relatively younger age, the annuity payments may be required to be made for a much longer period than the longest duration asset (i.e. Government bond) available. This would create a significant mismatch risk between assets and the liabilities during the annuity payout period.
- Given the 'participating' nature of this product, any transfers to shareholders (which are linked to declaration of bonuses to policyholders) can happen only during the deferment period (when the 'bonuses' are expected to be declared) and not during retirement / payout period. This means that any 'excessive' transfers to shareholders during the deferment period creates the risk of requiring significant capital infusions into the participating fund at the time of retirement / vesting, especially if the interest rates have lowered significantly as compared to those reflected in the guaranteed annuity rates offered.

b. Mortality / longevity risks

- Given the participating nature of the contract during deferment period, any mortality risk (i.e. actual experience being higher than expected) would be contained during deferment period. Also, if the level of guaranteed annuity rates is considered 'high', a 'higher-than-expected' mortality during deferment may help lower the cost of the guaranteed annuity payments during retirement / payout period.
- However, the Company will be subjected to significant longevity risks during retirement (especially for life annuities with no death benefit payable). This is more pronounced considering that the annuity rates are guaranteed at inception and it may be very difficult to estimate the improvement in life expectancy rates so far back from the period of retirement / payout.
- As the Company is offering (higher) 'impaired' life annuity rates, this too would add to the risk of potentially unexpected longevity of the 'impaired' life due to medical advancements etc. in the future.
- The product might not allow surrenders once the annuity payout commences. However, if it does, there may be a risk of selective withdrawals (i.e. more impaired lives surrendering their life annuities) impacting the overall longevity risk faced by the Company.

c. Expense risks

- Given the participating nature of the contract during deferment period, any expense risk (i.e. actual expenses being higher than expected) would be contained during deferment period.
- However, the guaranteed annuity rates at inception mean that the insurer may be subjected to significant expense risk during retirement / payout period, given the uncertainty about inflation rates in the long term.

d. Other risks

- Given the participating nature of the contract during deferment period, any persistency risk (i.e. higher or lower than expected lapse / surrender rates) may be contained during deferment period, especially if the Company's asset share framework re-cycles any lapse / surrender profits / losses back into the asset shares.
- However, a lower-than-expected lapse / surrender experience during deferment period may increase the cost of the guaranteed annuity payouts during retirement.

Non-participating annuity payable upon diagnosis of CI / inability to perform ADLs**a. Morbidity / CI risks**

- The Company may not have any experience in CI and incidence rates covering non-performance of ADLs. Any industry level experience data may also either not be available, or may not be relevant or may be out of date. Any experience data from reinsurers may also not be relevant if it is from other markets. This creates a significant risk of mis-pricing the product.
- Higher than expected incidences of CI or inability to perform ADLs pose a significant risk to the Company if the premium rates are non-reviewable and if the annuity rates are guaranteed given the non-participating nature of the product.
- Even if the premium rates are reviewable, there may be a delay in actually implementing a revision in premium rates in response to a worse than expected experience. There may also be caps imposed by the regulator on the maximum permissible increases in the premium rates. These too create risk for the Company.
- There is a risk of difference in interpretation of the legal / contractual definitions of CIs and ADLs, exclusions etc. between the Company and the policyholders, leading to the risk of higher-than-expected claims for the Company.

b. Longevity risks

- There is no data available pertaining to the life expectancy of annuitants following diagnosis of a CI or in a state that the policyholder is unable to perform ADLs. This creates a significant risk of mis-pricing of the product. Any improvement in the life expectancy, and particularly that following diagnosis of a CI or in a state that the policyholder is unable to perform ADLs, could lead to the Company paying claims for much longer than expected.

c. Market risks

- It is not clear if the annuity rates are guaranteed at inception of the contract. If they are, the Company is subjected to significant investment risk at the time of inception of the claim (i.e. upon CI or inability to perform ADLs) given that the interest rates at the time of vesting of annuity may be significantly lower than those reflected in the guaranteed annuity rates offered.
- Even if the annuity rates are not guaranteed, survival of the policyholder following a claim (e.g. diagnosis of CI) triggered at a relatively younger age may mean that the annuity payments may be required to be made for a much longer period than the longest duration asset (i.e. Government bond) available. This would create a significant mismatch risk between assets and the liabilities during the annuity payout period.

d. Other risks

- If the annuity rates are guaranteed at inception, this may mean that the insurer may be subjected to significant expense risk during the payout period, given the uncertainty about inflation rates in the long term.
- Given the possibility of mis-interpretation of legal / contractual terms pertaining to definitions of CI / ADLs etc., there is a significant mis-selling risk. Any delays at claims stage or claims investigations / underwriting resulting in repudiation of claims would create the risk of loss of reputation for the Company. This also creates a risk of regulatory intervention.

[2 marks for each of the main risks correctly identified and discussed in the context of the underlying benefits in the products, Max 14]

2. The difference in underwriting requirements between the two products is discussed below:

Participating deferred annuity with guaranteed annuity rates

- a. The main need catered to through this product is provision for retirement income. Given that the focus of this product is on 'financial savings', the underwriting requirements at inception will be focused on financial underwriting of the prospective customer. This may include:
- Know Your Customer (KYC) type verifications
 - Verification of age, address, bank account details etc.
 - Verification of income details to assess ability of the prospective customer to pay the premiums

- b. There is unlikely to be any medical underwriting required, unless the death benefit offered during the deferment period is very high and the deferment period is long. Nevertheless, the Company may seek basic information pertaining to the health of the policyholders, family medical history etc. in order to support any underwriting necessary at the time of retirement / vesting of the annuity.
- c. At the time of retirement / vesting of the annuity, given that the product offers separate rates for standard and impaired lives, the Company may need to verify the health status of the policyholder. For the policyholders seeking annuities based on the impaired annuity rates, a compulsory medical underwriting may be required to evidence the existence of impairment.

Non-participating annuity payable upon diagnosis of CI / inability to perform ADLs

- a. The main need catered to through this product is provision for income replacement upon diagnosis of CIs or upon policyholder's inability to perform the stated number of ADLs. Given that the focus of this product is on providing 'protection coverage through income replacement', the underwriting requirements at inception will be focused more on medical underwriting.
- b. However, the issue stage underwriting may be more to assess the risks related to likely incidences for diagnosis of CIs. Based on review of personal medical records / family medical history, it would be almost impossible to assess the risks related to incidences for policyholders' inability to perform stated number of ADLs. Thus, for the risk related to ADLs, the underwriting at inception may be only required to identify such policyholders who may already be showing signs of inability to perform ADLs. For this purpose, it may also be necessary to include cognitive testing as part of the underwriting process.
- c. Limited financial underwriting of the prospective customer may also be necessary. This may include:
 - Know Your Customer (KYC) type verifications
 - Verification of age (given that the risk of CI may be age dependent), address, bank account details etc.
 - Verification of income details to assess ability of the prospective customer to pay the premiums
- d. At the claims stage, the Company may also be required to carry out claims investigations / underwriting to identify any fraud, non-disclosures etc. This may include:
 - Assessment of medical records to establish diagnosis of the CIs covered in the contract
 - Assessment of medical practitioner's certificate (possibly from two different sources) to establish inability of the policyholder to perform the stated number of ADLs.

[6 marks for identifying and discussing the difference in the underwriting standards between the two products based on the differences in product features.]

[Max 20]

iii)

1. The possible reasons for lower surrender values in the proposed participating deferred annuity product as compared to the other participating endowments offered by the Company are:
 - a. **Deterrence** - Given the 'retirement income provision' focus of the deferred annuity plan, the Company may have consciously provided lower surrender values in this plan in order to deter the policyholders from surrendering their policies which are meant as a source of retirement income.
 - b. **Recovery of guarantee costs** - The guaranteed annuity rates offered in the deferred annuity plan means that the Company may wish to recover the cost of providing such a guarantee through implicit charges – one of them being through offering lower surrender values.
 - c. **Higher profitability** - The overall expected profitability level of the deferred annuity plan may be higher than the other participating endowments considering the guarantee risks undertaken by the Company. Given this, the Company may have offered lower overall benefits (e.g. lower surrender values) for the same amount of premiums payable in the two products.

- d. **Difference in timing of pricing of the products** - The other participating endowment products may have been priced in a different economic / business environment than that applicable today.
- For example, if the interest rates have come down since the pricing of the participating endowments and if the Company has not revised the surrender values to reflect the lower interest rate environment, the surrender values in the other participating endowments may continue to be higher than those offered in the proposed deferred annuity product.

- The Company's cost structure may also be different today as compared to that applicable at the time of pricing of the other participating endowments. Although the overall commission rates in a deferred annuity product may be lower than those in other participating endowments, if the Company's overall cost structure has increased, this may result in lower surrender values for the proposed deferred annuity product.

[1 mark for giving each of the possible reasons for low surrender values provided with suitable explanation, Max 4]

2. The possible avenues available to the Company to offer competitive surrender value scales in the proposed deferred annuity product are:

- Accept lower profitability** – The Company may be able to improve the surrender values if it is prepared to accept a lower level of profitability. The profitability may not reduce significantly anyway, if the actual surrenders remain low considering that the main focus of the product is retirement provisions.
- Re-align benefit structure** – If a lower level of profitability is not acceptable, the Company may explore re-alignment of the benefit structure – e.g. it may be able to improve the surrender values by lowering other benefits (e.g. death benefits / vesting benefits etc.)
- Re-visit the cost structure in the deferred annuity plan** - For example, the Company may explore if a lower level of distributor compensation is acceptable. This may improve the projected asset shares in the product, allowing the Company to offer higher surrender values which are expected to reflect the asset shares.
- Re-visit the investment strategy applicable** - Considering that the product is meant for retirement provision, the Company may re-visit the investment strategy during the deferment period to explore if higher investment returns can be generated (e.g. through increased equity type investments). This may improve the projected asset shares in the product, allowing the Company to offer higher surrender values.

[1 mark for giving potential avenues available to improve surrender values provided with suitable explanation, Max 4]

[Max 8]

iv)

The two main issues experienced by the Company – a decline in new business volumes and high level of surrenders from existing policyholders – may require different responses / interventions by the Company.

The possible steps that may be taken by the Company to make the product more attractive to the customers are discussed below:

For existing policyholders

- Communication** – The Company may wish to reach out to the existing policyholders and highlight the key features of their policies to deter them from surrendering. These include:
 - How the product is meant for the long-term with the focus being on provision for retirement income;
 - How the policyholders would get the benefit of any higher investment returns in the future (i.e. if the high interest rates currently experienced sustain) through higher bonuses in the future;
 - Issuing an updated benefit illustration highlighting the enhanced level of future benefits and the resulting annuity payment;

- Highlighting the value of the guaranteed annuity rates – something that no other financial product provides;
- Highlighting the tax benefits available through the product and possibility of loss of such benefits if the policy were to be surrendered;
- How the policyholders may 'lose' by surrendering their policies prematurely;
- Making them aware of the reduced paid-up (RPU) feature offered in their policies;
- Making them aware of the feature of availability of a loan against their policies, should they require liquidity etc.

- b. **Downwards revision in surrender value scales** – The increase in short-medium term interest rates would mean that the market values of the assets would now be lower. This may warrant the surrender values to be lowered, unless the Company re-cycles any market value gains or losses on the lapse / surrender profits or losses back into the asset shares of the remaining policyholders. This may also act as a deterrent from surrendering the policies.

However, this may need to be considered carefully as any downward revision in the surrender value scales may increase the perception of 'lack of value for money' in the policy amongst the policyholders, risking a further increase in the surrender rates experienced.

- c. **Short term offers** – Explore if certain short-term offers can be made (subject to regulatory approvals) to ensure that the policyholders do not surrender their policies. For example, these may include:
- Enhancement of policy values through a special one-off bonus for policyholders who complete say three policy years;
 - Reduction in interest rates charged on policy loans.

For potential new customers

- a. **Revising the product feature to make it more attractive** –
- Enhance the level of benefits offered by, for example, by reducing the distribution costs, lowering underwriting costs, offering the product through online / direct distribution channel platforms etc.
 - Re-structuring the benefits in the product – e.g. increasing the retirement / vesting benefits by lowering surrender benefits (if policyholders value the retirement benefits more) or vice versa.
 - If the guaranteed annuity rates feature is valued by the policyholders, explore if the level of guarantee offered can be further enhanced. This may be feasible, if the interest rates have increased for upto 10 years. Alternatively, if the guarantees are not valued as much by the policyholders, the level of guarantees offered can be either lowered or completely eliminated – allowing the Company to significantly lower the cost of guarantees offered and resulting in a potentially significant enhancement in the overall benefits of the product.
- b. **Repricing of the product given the changed interest rate environment**
- If the changes in the interest rate environment are considered to be permanent, the Company may want to re-price the product. This may make the product more attractive to the customer.
 - If the changes in the interest rate environment are considered to be temporary, the Company may be able to reflect the same in the declared bonus rates. Hence, re-pricing may not be required. Nonetheless, to address the policyholder perception issue, the Company may consider re-pricing the product anyway.
- c. **New product design on unit-linked or index-linked platforms** - The Company may explore introducing the deferred annuity product on a unit-linked or an index-linked platform. This may have the following benefits:
- In the unit-linked design, a choice of funds (which may include 100% equity-oriented funds) offered during the deferment period would allow the policyholder to benefit from potentially higher policy values at retirement / vesting.
 - In the index-linked design, although the policyholder may not have much flexibility in selecting the funds to invest in during the deferment period, but the interest credited in the policy may be directly

linked to the market returns on the specified index, thereby making it much more transparent for the policyholder.

- Explicit charging structure in these two product designs may make it transparent to the policyholder, thereby increasing the attractiveness of the product.
- Given that the surrender charges may be subject to regulatory limits, the overall surrender values in the initial years may be higher. If low level of surrender values in the existing product design is perceived to be an issue, these two product constructs may help address the same.

[2 marks for each of the appropriate steps to enhance the attractiveness of the product suitably discussed, **Max 12]**

[50 Marks]
